Matthew F. Panichello

Curriculum Vitae
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CONTACT

318 Campus Drive
Stanford, CA, 94305

mfp2@stanford.edu
(410) 707-2814
https://panichem.github.io/
google scholar

EDUCATION

PhD | Princeton University 2020
Neuroscience

BS | Boston College 2011
Psychology, Philosophy *summa cum laude*

RESEARCH POSITIONS

Research Scientist, Stanford University and HHMI Jan 2026–
Advisor: Tirin Moore

Neuroscientist, Starfish Neuroscience Jan–Dec 2025

Postdoctoral Fellow, Stanford University 2021–2024
Advisor: Tirin Moore

Graduate Student, Princeton University 2013–2020
Advisors: Tim Buschman & Nick Turk-Browne

Research Coordinator, Massachusetts General Hospital (MGH) 2011–2013
Advisor: Moshe Bar

Research Assistant, MGH & Boston College 2008–2011
Advisor: Lisa Feldman Barrett

Advanced Study Grant Fellow, Columbia University 2009
Advisor: Tor Wager

PUBLICATIONS

Kim, D., Panichello, M.F., & Moore, T. (2026). Condition-dependent noise correlations without condition-dependent spike counts. *bioRxiv*.

Muralidharan, S., Leng, C., Orts, L., Trepka, E., Zhu, S., Panichello, M.F., Jonikaitis, D., Lanfranchi, F., Pennington, J., Pachitariu, M., & Moore, T. (2026). Live spike sorting of large-scale neural recordings. *bioRxiv*.

Panichello, M.F., Jonikaitis, D., Oh, J., Zhu, S., Trepka, E.B., & Moore, T. (2024). Intermittent rate coding and cue-specific ensembles support working memory. *Nature*, 636, 422-429.

Alleman, M., Panichello, M.F., Buschman, T.J., & Johnston, W.J. (2024). The neural basis of swap errors in working memory. *PNAS*, 121(3).

Apostel, A., Panichello, M.F., Buschman, T.J., & Rose, J. (2023). Corvids optimize working memory by categorizing continuous stimuli. *Communications Biology*, 6(1), 1122.

Panichello, M.F., & Buschman, T.J. (2021). Shared mechanisms underlie the control of working memory and attention. *Nature, 592*, 601-605.

Panichello, M.F., & Turk-Browne, N. (2021). Behavioral and neural fusion of expectation with sensation. *Journal of Cognitive Neuroscience, 33*(5), 814-825.

Yu, Q., Panichello, M.F., Cai, Y., Postle, B.R., & Buschman, T.J. (2020). Delay-period activity in frontal, parietal, and occipital cortex tracks noise and biases in visual working memory. *PLOS Biology, 18*(9).

Panichello, M.F., DePasquale, B., Pillow, J.W., & Buschman, T.J. (2019). Error-correcting dynamics in visual working memory. *Nature Communications*, *10*(1).

Panichello, M.F., Kveraga, K., Chaumon, M., Bar, M., & Barrett, L.F. (2017). Internal valence modulates the speed of object recognition. *Scientific Reports*, *7*, 361.

Panichello, M.F., Cheung, O.S., & Bar, M. (2013). Predictive feedback and conscious visual experience. *Frontiers in Psychology*, *3*, 620.

INVITED TALKS

Humboldt-Princeton Symposium, Primate and Human Working Memory	2027 (upcoming)
UC Berkeley, Cognitive Neuroscience Colloquium	2026 (upcoming)
Princeton University, Nonhuman Primate Symposium	2026
Cosyne, Mechanisms of Working Memory Workshop	2025
University of Tübingen, Neural Subspace Workshop	2024

UC Berkeley, Knight Lab Meeting	2023
UC Davis, Vision Journal Club	2021
Ruhr University Bochum, Rose Lab Meeting	2021
UC Riverside, Zagha Lab Meeting	2021

FELLOWSHIPS AND GRANTS

National Defense Science and Engineering Graduate Fellowship	2015–2018
McDonnell Fellow in Neuroscience, *Princeton University*	2014–2015
Undergraduate Research Fellowship, *Boston College*	2008–2010
Advanced Study Grant, *Boston College*	2009

TEACHING

NSUR 249: Experimental Immersion in Neuroscience, Guest Lecturer	2021
PSY 255: Cognitive Psychology, Preceptor	2015
NEU 258: Fundamentals of Neuroscience, Preceptor	2014
Foundation Academy Charter School, Volunteer Tutor	2013–2014
Let's Get Ready (free SAT prep), Volunteer Instructor	2008

STUDENTS ADVISED

Doyeon Kim, master's thesis (Stanford University)	2026
Tony Gao, undergraduate summer study (Stanford University)	2026
Rio Naka, undergraduate research (Stanford University)	2024
Jin Oh, post-baccalaureate research (Stanford University)	2021–2023
David Mitchell, undergraduate thesis (Princeton University)	2017–2018
Timothy Baum, undergraduate summer study (Princeton University)	2017
Christian Wawrzonek, undergraduate thesis (Princeton University)	2015–2016
David Waldinger, undergraduate summer study (MGH)	2012

SERVICE & OUTREACH

Stanford Community College Outreach Program	2024
Princeton Neuroscience Curriculum Committee	2014–2015
Princeton Neuroscience Graduate Student Committee, Chair	2014–2015
Psi Chi Psychology Honors Society, Boston College	2008–2011
President	2010–2011

AD-HOC REVIEWER

Nature Communications, Nature Human Behavior, PNAS, eLife, Journal of Neuroscience, Cerebral Cortex, Scientific Reports, Cortex, Psychonomic Bulletin & Review